COURSE CODE: PY221IC
USN

RV COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU
II Semester B.E. July-2026 Examinations

DEPARTMENT OF PHYSICS

Model Question Paper

COURSE TITLE: Quantum Physics for Engineers
(2025 SCHEME)

Time: 03 Hours Maximum Marks: 100

Instructions to candidates:

1) Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2) Answer FIVE full questions from Part B. In Part B question numbers 2 & 3 are compulsory. Answer any one full question from 4 or 5, 6 or 7 and 8 or 9.

PART A

Q. No	Questions	Marks	CO	BT
1.1	de Broglie waves are not electromagnetic waves, justify.	01	1	1
1.2	What is the physical significance of wave function?	01	1	1
1.3	If the Fermi level for an intrinsic semiconductor is at 3 eV, what is the probability of occupation of electrons at energy level 3.5 eV at zero kelvin?	01	2	2
1.4	What is a Hall effect in metals?	01	1	1
1.5	What is the value of magnetic susceptibility of a type-I superconductor?	01	1	1
1.6	What are cooper pairs in superconductors?	01	1	1
1.7	In which bias does avalanche photo diode and single-photon avalanche diode work?	01	1	1
1.8	What is stimulated emission in laser?	01	1	1
1.9	Which principle allows a qubit to exist in multiple states simultaneously?	01	1	1
1.10	Which quantum gate is also known as the NOT gate?	01	1	1

PART B

Q.No	Unit 1	Marks	CO	BT
2a	Using the time independent Schrodinger's wave equation, find the solution for a particle in an infinite potential well of width 'a'. Hence obtain normalized wave function.	08	2	2
2b	Explain de Broglie hypothesis of matter waves and mention the properties of matter waves.	06	2	2
2c	Calculate the lifetime of an excited state in an atomic system, given that the emitted spectral line has a wavelength of 6000 Å and a spectral line width ($\Delta\lambda$) of 10^{-4} Å.	04	3	3
	Unit 2			
3a	Prove that the Fermi level of an intrinsic semiconductor lies in the middle of the band gap at zero kelvin. Sketch and explain the variation of carrier concentration with temperature in an n-type semiconductor.	08	2	2
3b	Write any two assumptions of Quantum free electron theory. Arrive at an expression for carrier concentration in metals at zero kelvin.	06	1	1
3c	A copper strip 2 cm wide and 1 mm thick is placed in a magnetic field of 1.5 Wb m^{-2}. If a current of 200 A is set up in the strip, calculate Hall voltage that appears across the strip. Assume $R_H = -6\times10^{-7}\ m^3C^{-1}$.	04	3	3
	Unit 3			
4a	Explain the Bardeen-Cooper-Schrieffer (BCS) theory of superconductivity, including its key mechanisms and the concept of phase coherent state.	08	2	1
4b	Explain the working of a DC SQUID device with relevant diagrams.	06	2	2
4c	A type-I superconducting cylindrical wire has radius 0.5 µm and critical magnetic field 0.05 T. Calculate the critical current. If the radius becomes 1 mm, find the new critical current.	04	3	3
	OR			
5a	Explain Type-I and Type-II superconductors with neat sketches, highlighting their key differences in magnetic behavior and critical fields.	08	2	1
5b	Explain AC Josephson effect with a neat diagram. Mention any two applications of superconductor.	06	2	1
5c	A Josephson junction has a 2.5 µV DC bias, causing microwave emission via the AC Josephson effect. Find the emission frequency. Calculate the corresponding photon energy. If the voltage doubles, explain the wavelength change qualitatively.	04	3	3
	Unit 4			
6a	Explain the construction and working of a Semiconductor Diode LASER with the help of energy band diagram.	08	1	1

6b	Explain the working of Mach-Zehnder Interferometer.	06	2	2
6c	The ratio of population of two energy levels out of which one corresponds to metastable state is 1.059×10^{-30}. Find the wavelength of light emitted at 330 K.	04	3	3
	OR			
7a	Discuss the different attenuation mechanism on an optical fiber.	08	1	1
7b	Derive the expression for energy densities of incident radiation in terms of Einstein's coefficients.	06	2	2
7c	An optical fiber in air has numerical aperture of 0.20 and a cladding refractive index of 1.59. Determine the acceptance angle for the fiber in water which has a refractive index of 1.33.	04	3	3
	Unit 5			
8a	a) With a neat labelled diagram explain the construction and working of the single photon interference experiment, without the half wave plate. What is the inference of this experiment ?	08	2	3
8b	Represent a general Qubit state with the help of a Bloch sphere. Explain the action of the following gates on an arbitrary point on the Bloch sphere: a) Pauli X gate, b) Hadamard gate	06	2	3
8c	Calculate the action of Pauli X gate followed by Hadamard gate on the state $\|0\rangle$	04	3	3
	OR			
9a	a) Write the matrix representation of a) Hadamard Gate, b) CNOT gate. Explain their action in Dirac Notation. Write down their truth tables.	08	1	3
9b	Explain three differences between classical and quantum computing.	06	2	3
9c	A qubit is in the state $\| \psi\rangle = \frac{1}{\sqrt{N}} \begin{bmatrix} 3 \\ i\sqrt{7} \end{bmatrix}$ a) Verify that $\|\psi\rangle$is a normalized state. If not, calculate the normalized coefficient.	04	3	3

Signature of the Scrutinizer:

Name

Signature of the Chairman:

Name